LOCKED

Targets, data and field tools | 29 pages | US Letter, true scale



TARGETS

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FIELD KIT

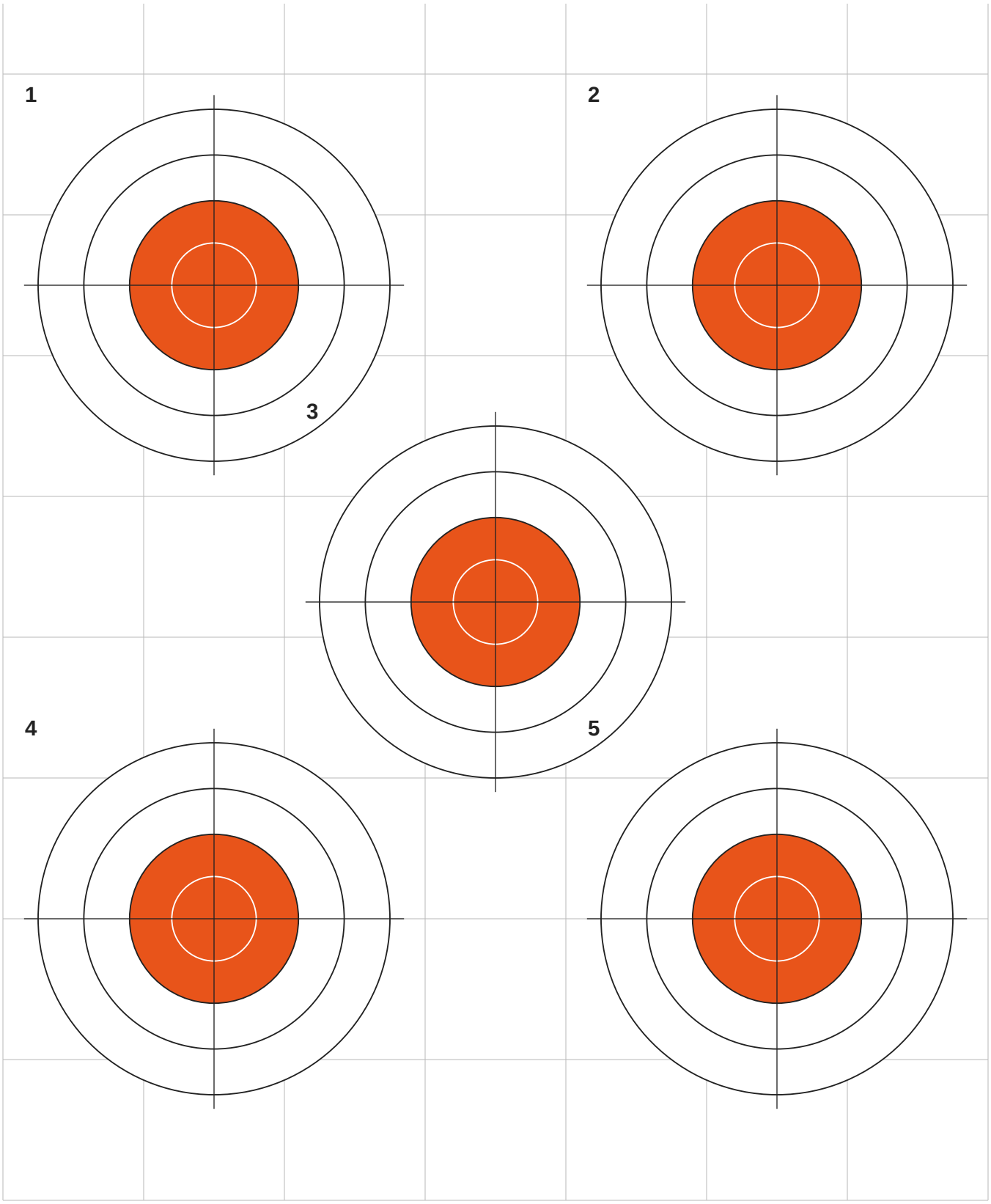
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To print one target, print just its page number. Always print at 100% / Actual Size so the scales stay true.
Tap any row to jump to that page. Laminate the cards and checklists for field use.

FIVE-BULL SIGHT-IN

T-01

General sight-in. Five 2.5 in. bulls on a 1 in. grid. Pistol 7-15 yd, rifle 25-50 yd.



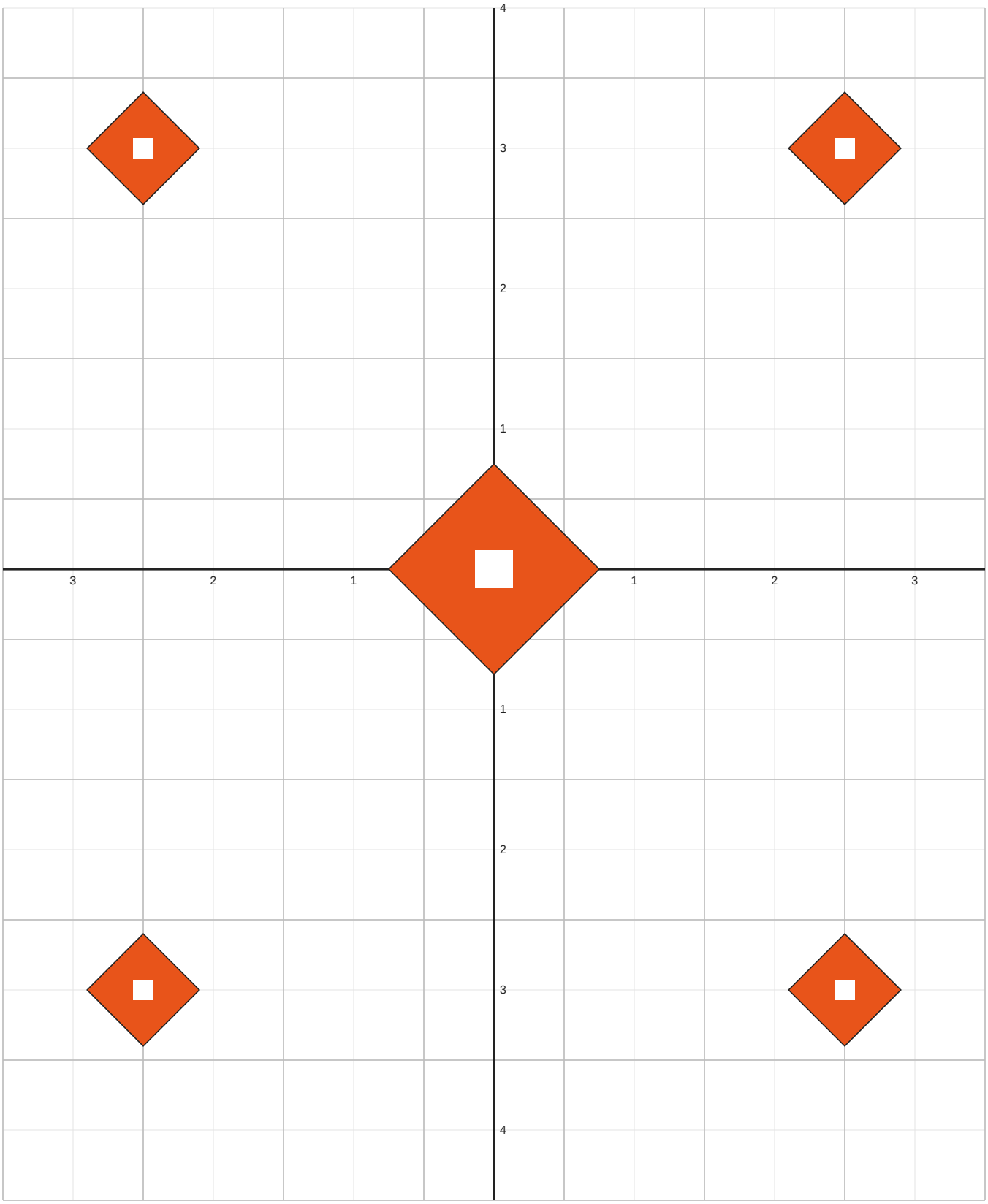
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

100-YARD ZERO GRID

T-02

1 in. squares. At 100 yd each square is about 1 MOA (1.047 in.). Center diamond for zero, corners for confirmation.



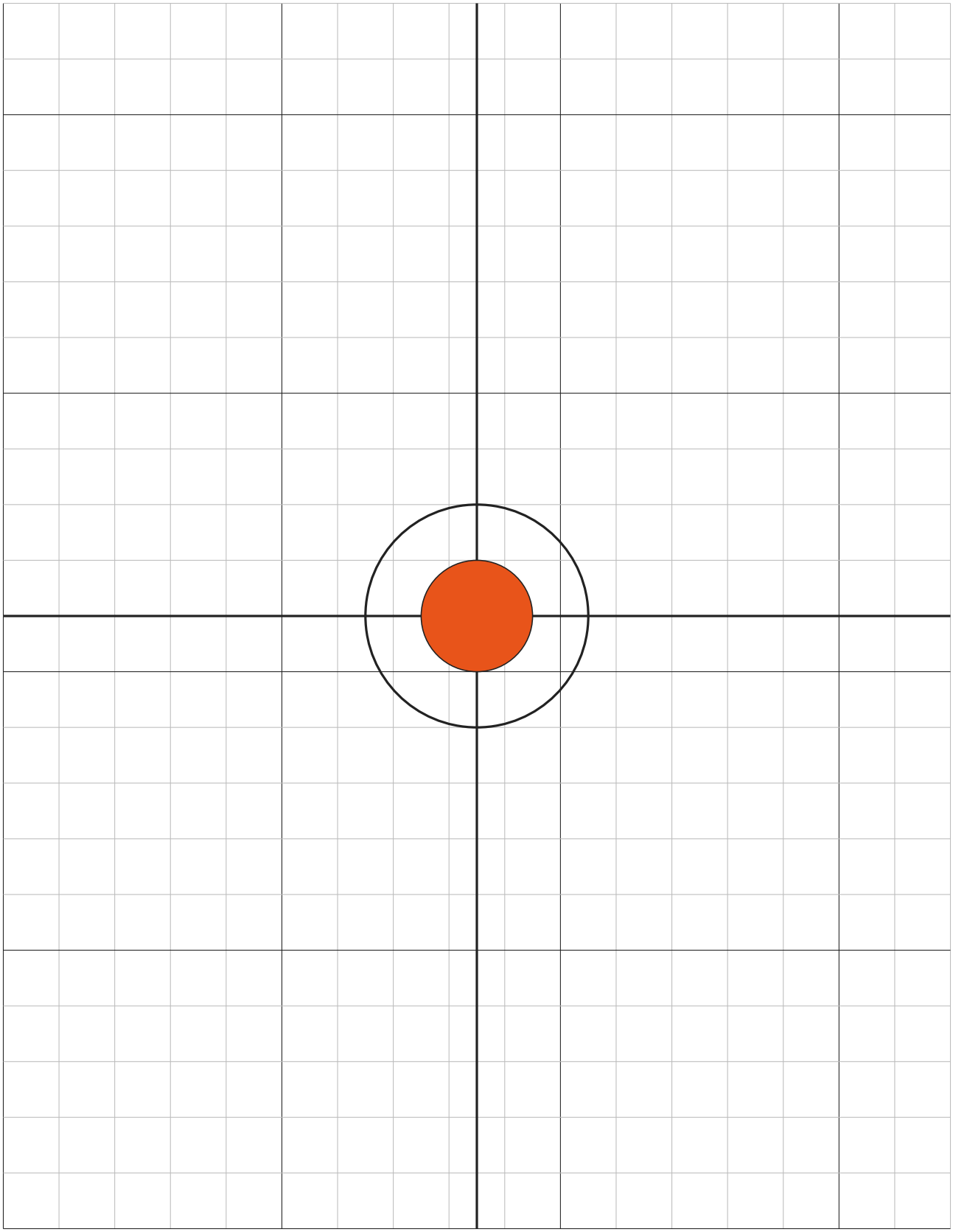
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

25 M METRIC ZERO

T-03

1 cm squares. At 25 m each square = 0.4 mil (about 1.4 MOA). Move impacts to the center circle.



Major lines every 5 cm. Aim at center; group should land inside the 4 cm circle.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

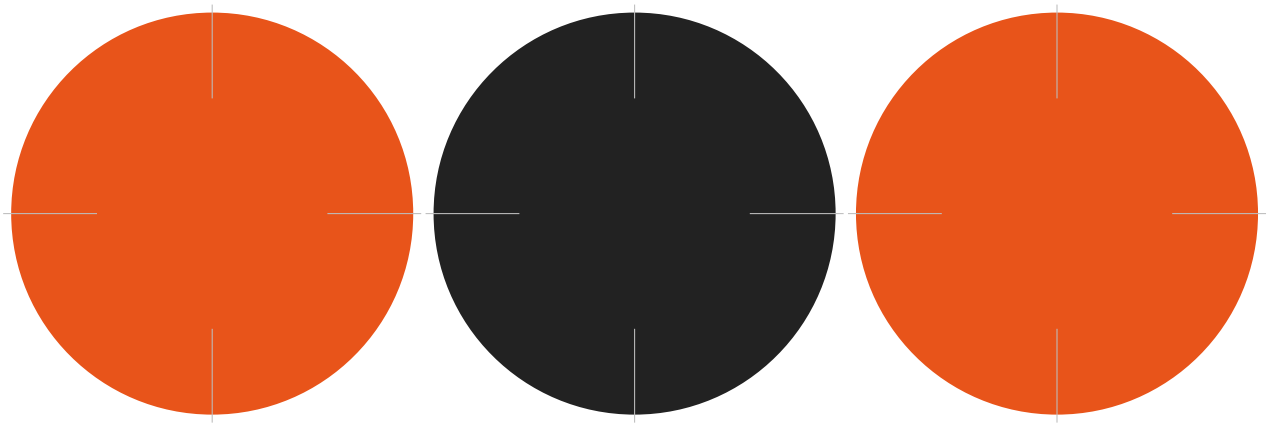
= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

MOA DOT LADDER

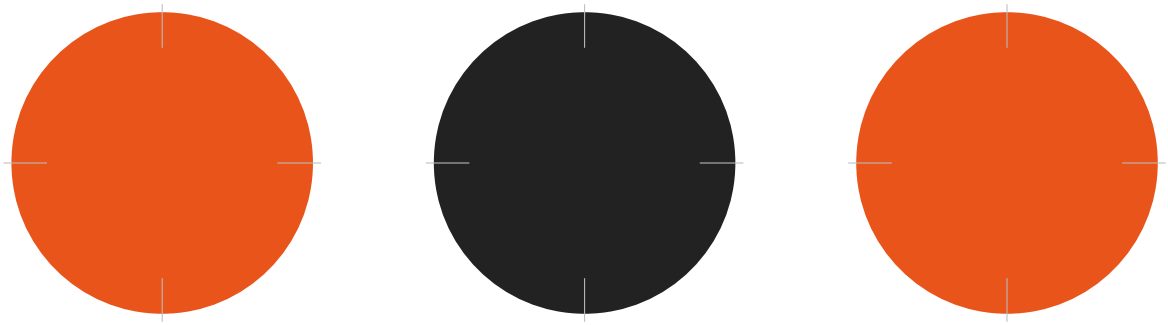
T-04

Dot sizes in true MOA at 100 yd. Rows: 2, 1.5, 1 and 0.5 MOA. Shoot each row smaller as you tighten up.

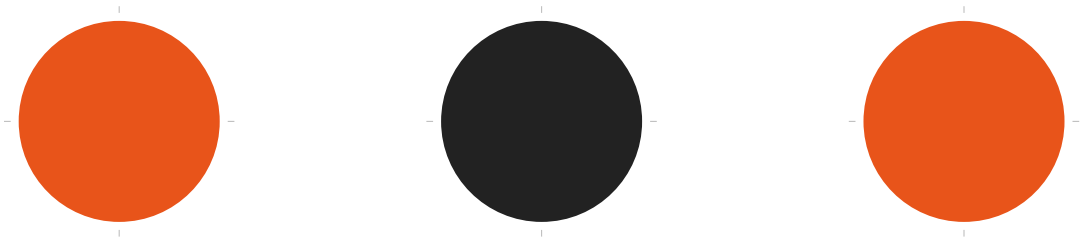
2 MOA



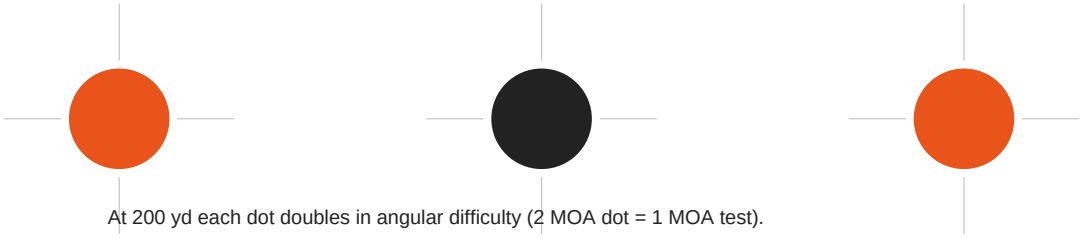
1.5 MOA



1 MOA



0.5 MOA



At 200 yd each dot doubles in angular difficulty (2 MOA dot = 1 MOA test).

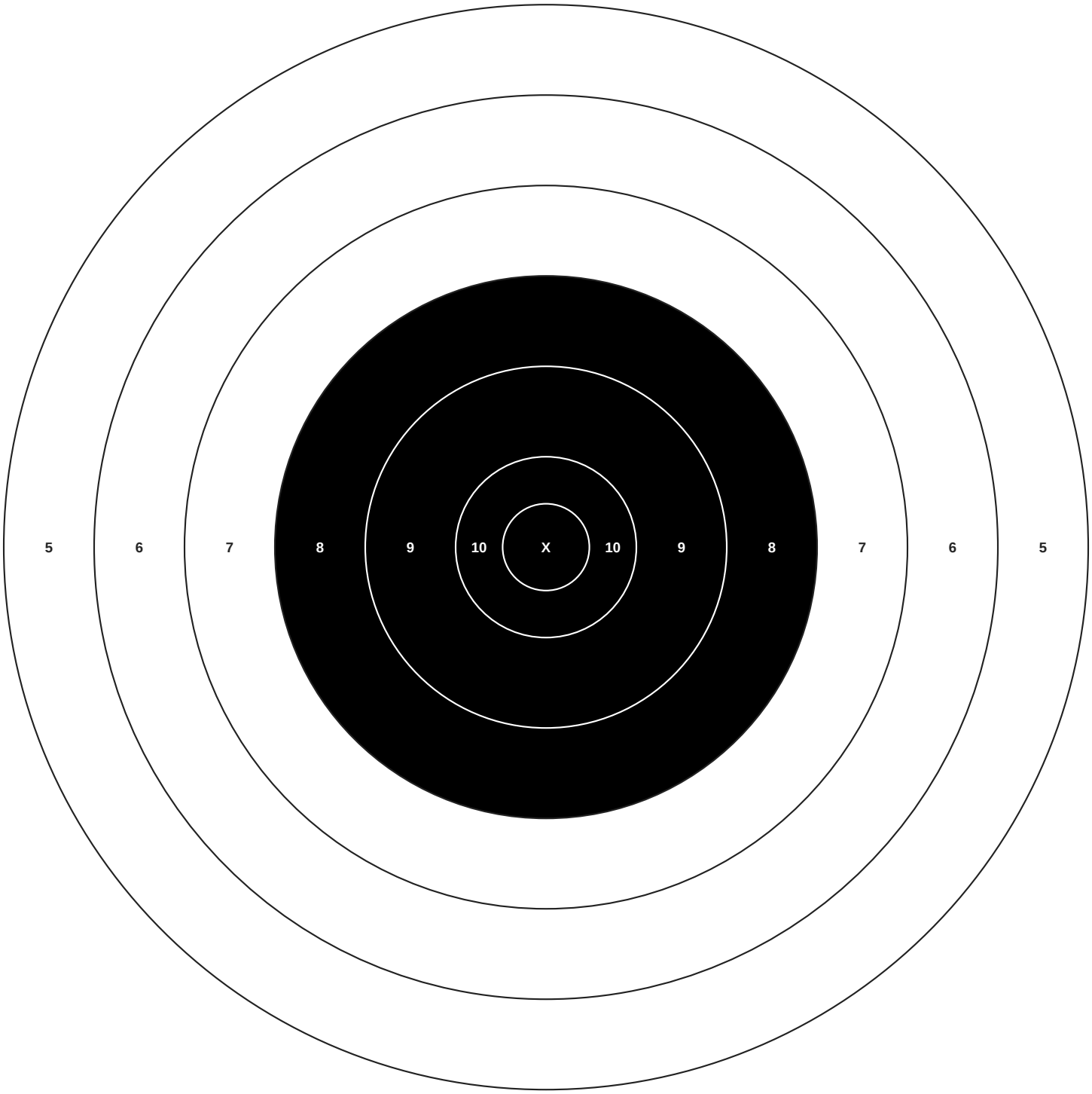
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

PRECISION BULLSEYE

T-05

Slow-fire scoring target. Rings 10 to 5 plus X ring. Score each shot by the highest ring it touches.



1	2	3	4	5	6	7	8	9	10	TOTAL

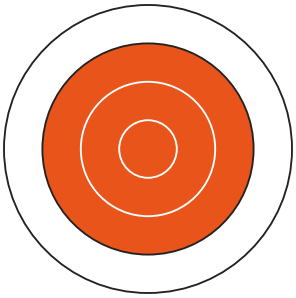
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

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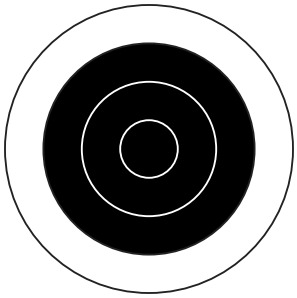
RIMFIRE 12-BULL

T-06

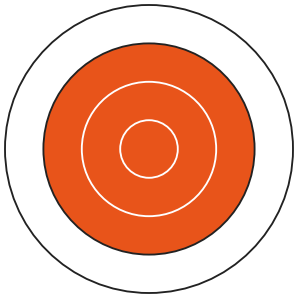
Ten 1.5 in. record bulls plus two sighters. Built for 50 ft / 25 yd rimfire and air rifle practice.



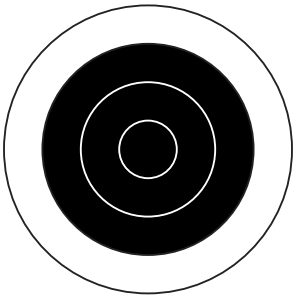
SIGHTER



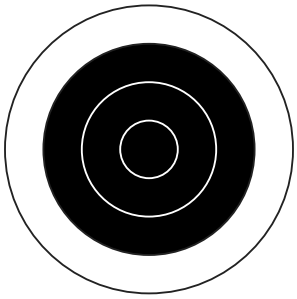
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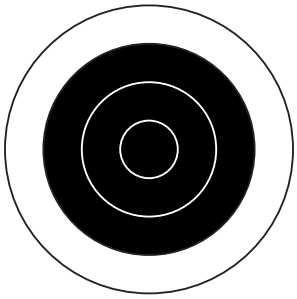
SIGHTER



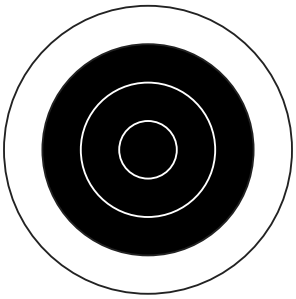
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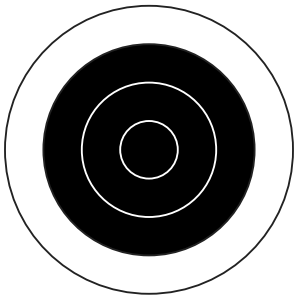
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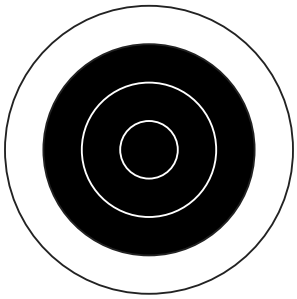
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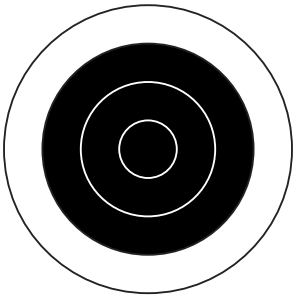
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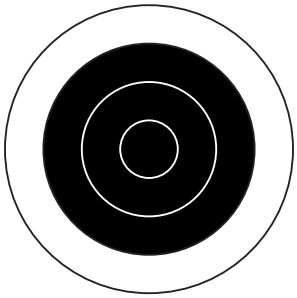
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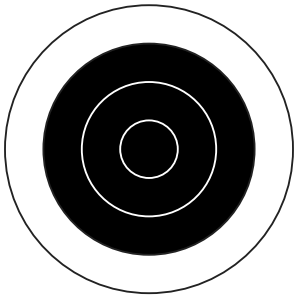
7



8



9



10

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

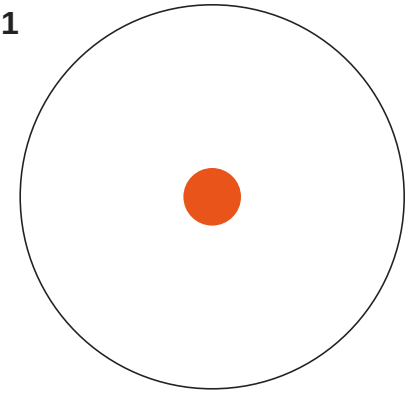
= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

TEN-DOT DRILL

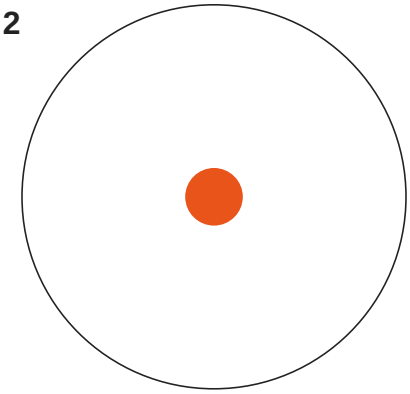
T-07

2 in. and 1 in. dots. Run each numbered string in order. Any miss on a string means you rerun it.

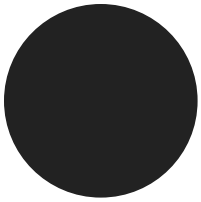
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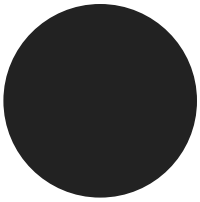
2



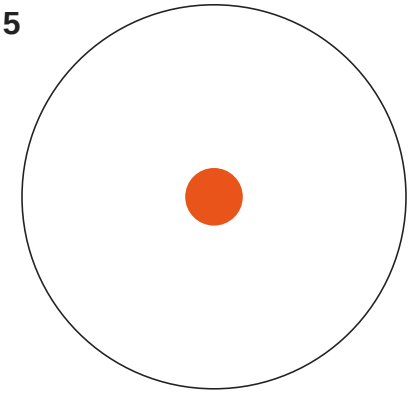
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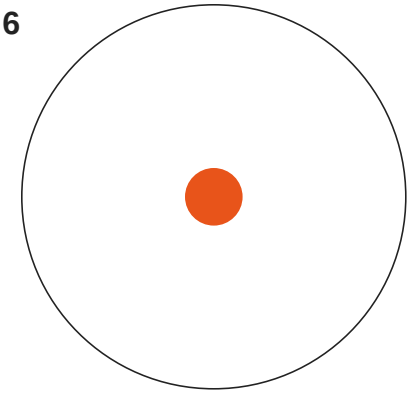
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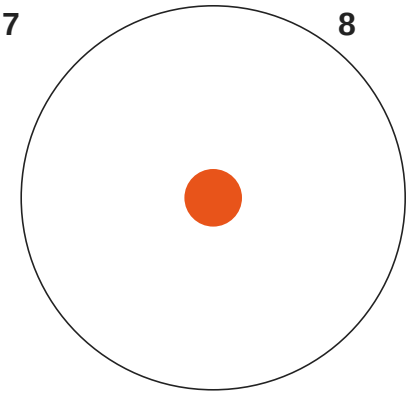
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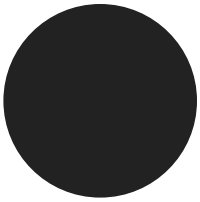
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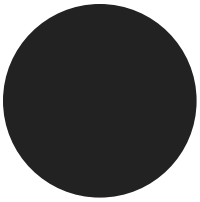
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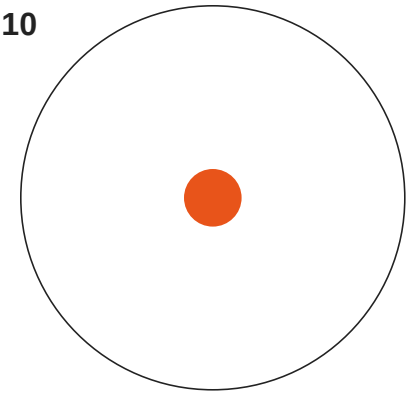
8



9



10



STRINGS (start at 3 yd, back up once you run it clean)

1. 5 shots, slow fire
2. Draw, 1 shot. x5
3. 5 shots, slow fire
4. Draw, 1 shot. x5
5. Draw, 2 shots. x3
6. Strong hand only, 5
7. Weak hand only, 5
8. 1 shot each on 8 then 9, x4
9. (see 8)
10. Reload drill: 1, reload, 1. x2

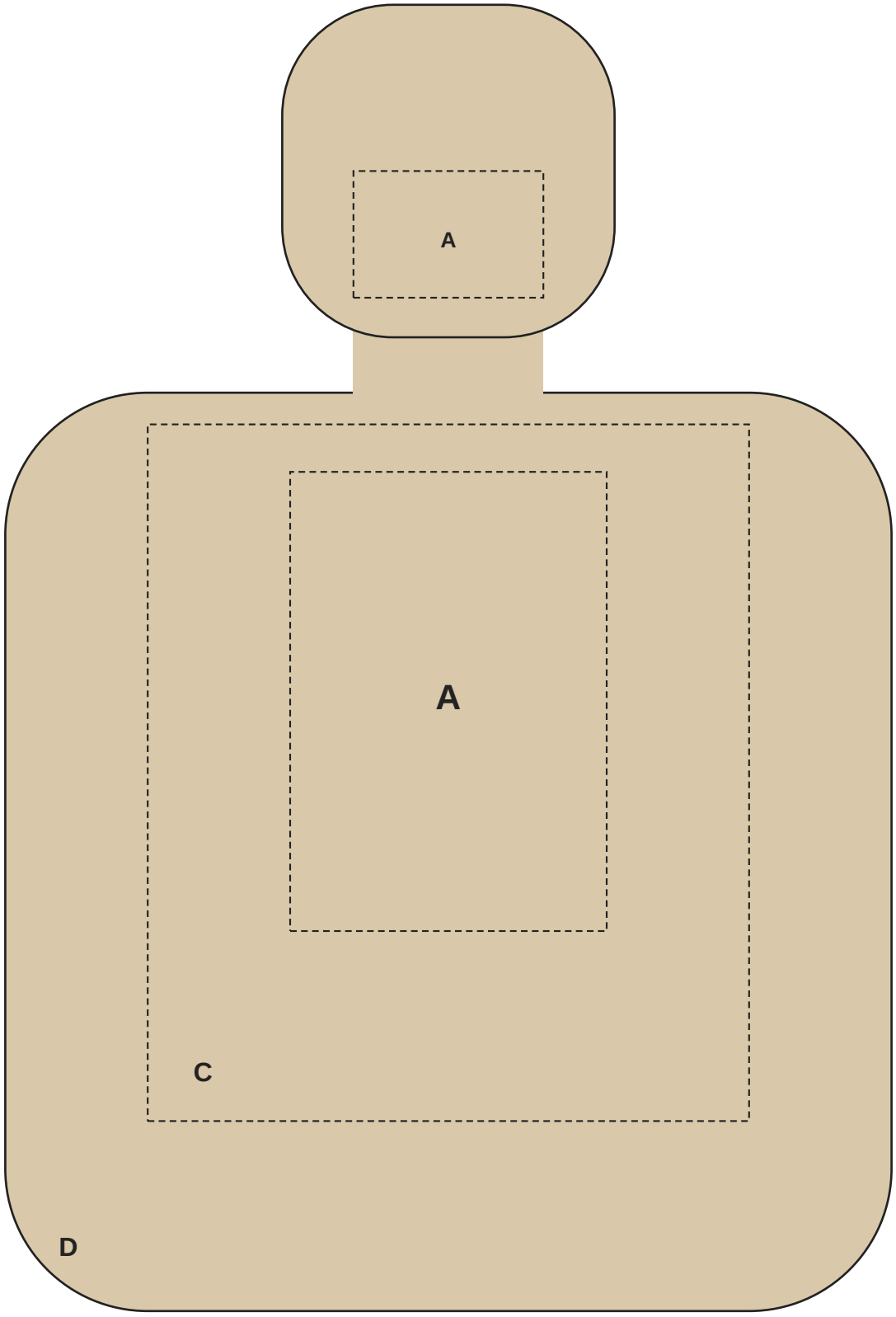
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

REDUCED SILHOUETTE

T-08

Generic torso and head with scoring zones. Reduced size: shooting it at 10 yd looks like a full-size target at about 30 yd.

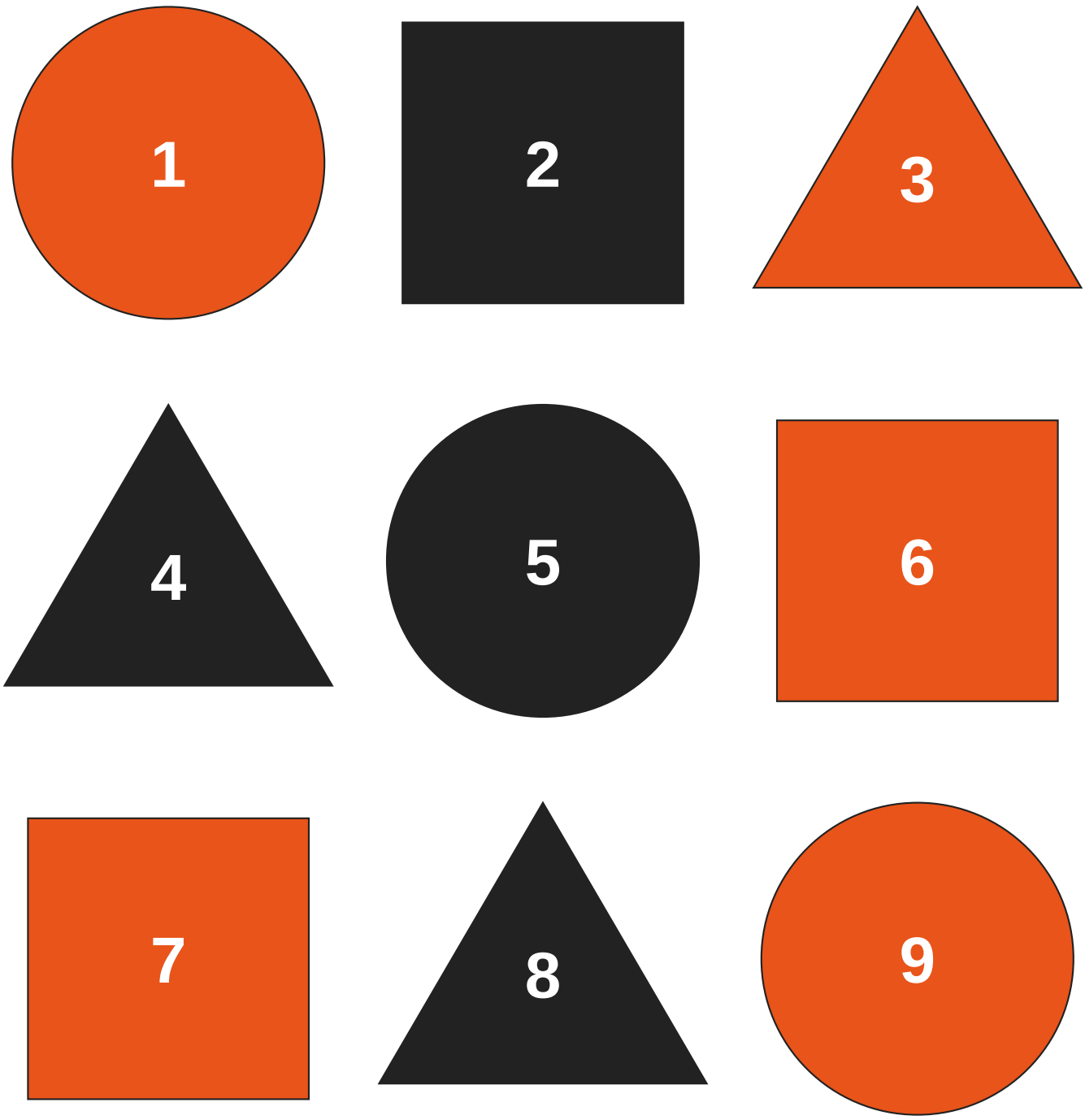


SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

SHAPES & NUMBERS

Call-out drill. A partner calls a shape, a color, or a number and you engage only that target. Builds target ID under time.

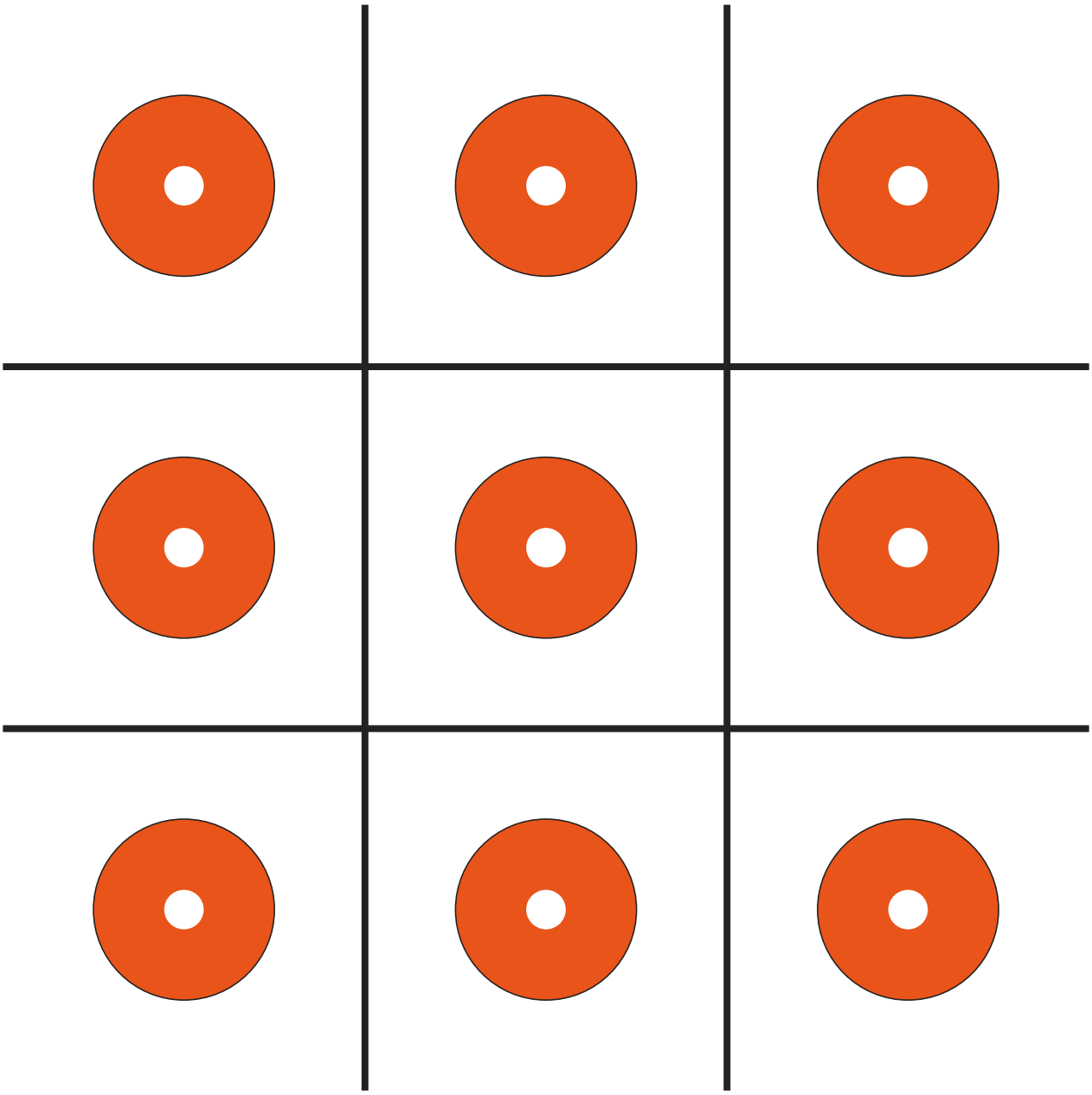


SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

TIC-TAC-TOE

Two shooters, one target. Hit a square to claim it with your color of marker. First to three in a row wins.



Tip: use colored target pasters or a marker to claim squares between strings.

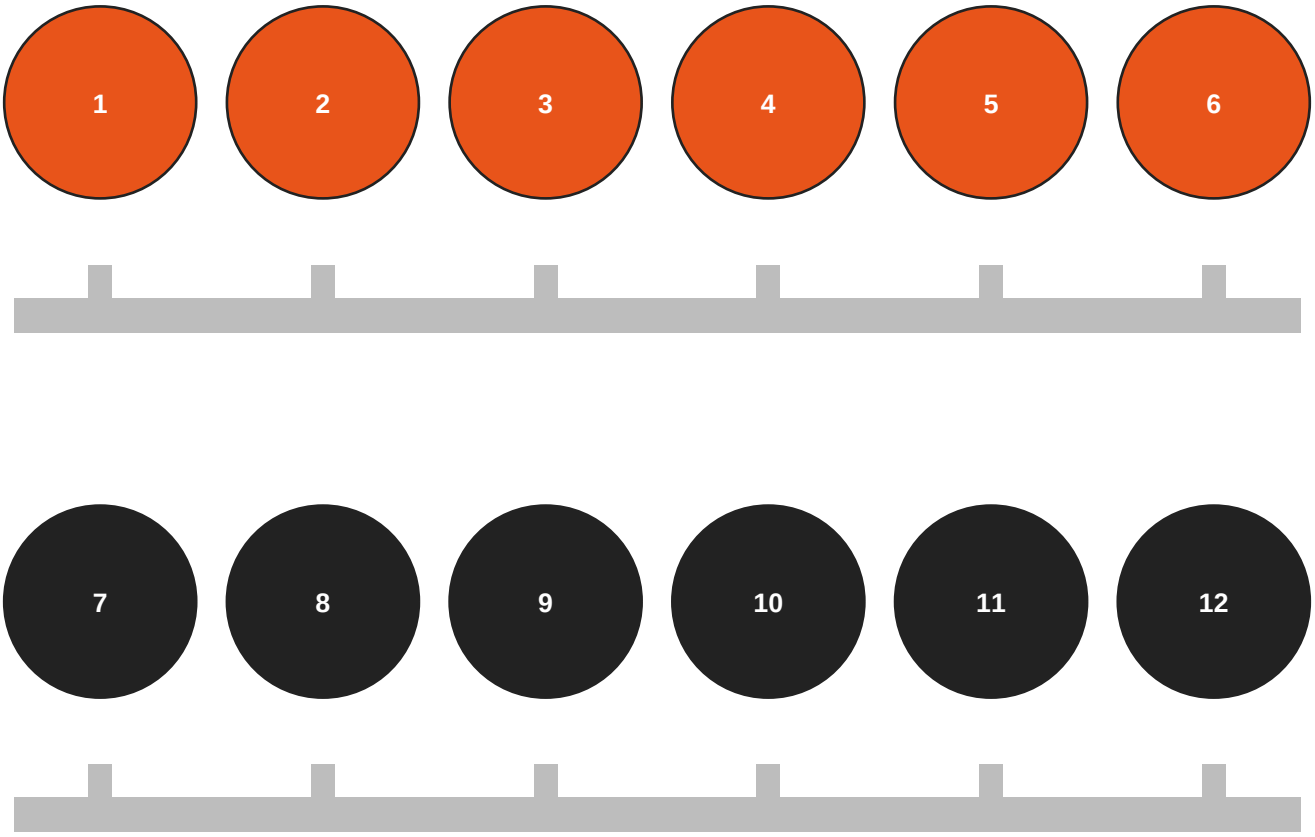
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

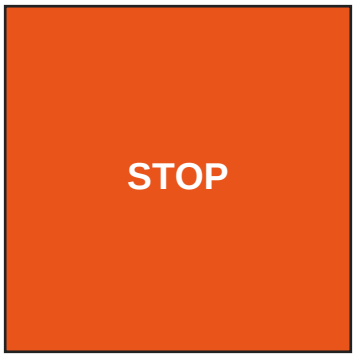
PLATE RACK

T-11

Two racks of six 1 in. plates plus a stop plate. At 10 yd each plate matches a 4 in. steel plate at 40 yd.



Clear a rack left to right, then hit STOP. Log your time.



SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

ERROR DIAGNOSIS

For a right-handed handgun shooter (mirror it if left-handed). Fire 10 slow shots at the center, then read where they cluster.



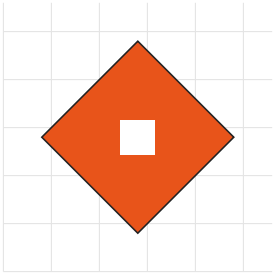
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

LOAD DEVELOPMENT

Ten 1 in. diamonds for a ladder or OCW test. Write the charge and seating depth under each.

LOAD 1

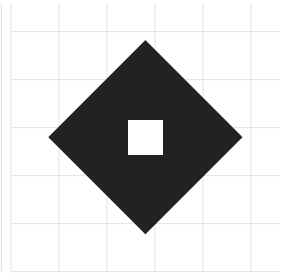


CHG _____

OAL _____

VEL _____

LOAD 2

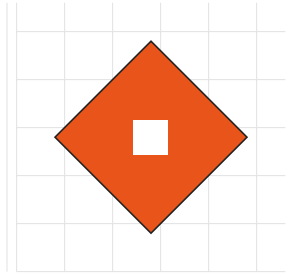


CHG _____

OAL _____

VEL _____

LOAD 3

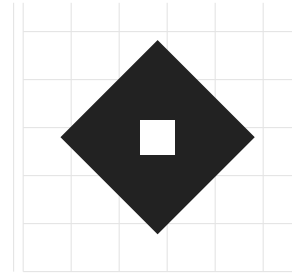


CHG _____

OAL _____

VEL _____

LOAD 4

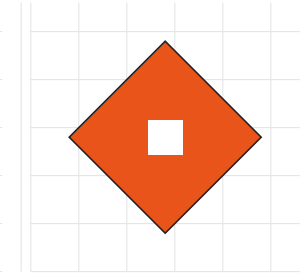


CHG _____

OAL _____

VEL _____

LOAD 5

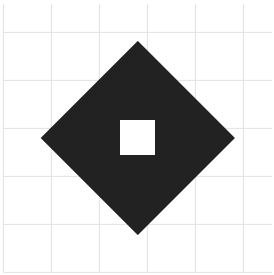


CHG _____

OAL _____

VEL _____

LOAD 6

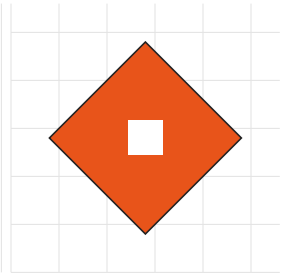


CHG _____

OAL _____

VEL _____

LOAD 7

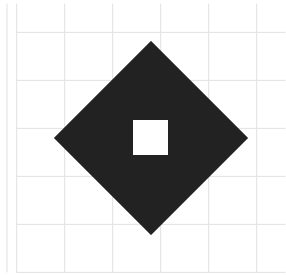


CHG _____

OAL _____

VEL _____

LOAD 8

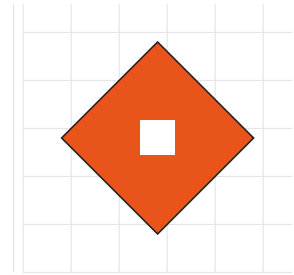


CHG _____

OAL _____

VEL _____

LOAD 9

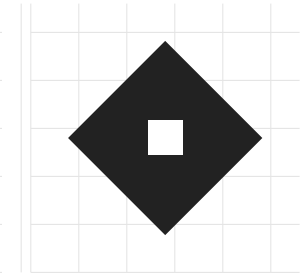


CHG _____

OAL _____

VEL _____

LOAD 10



CHG _____

OAL _____

VEL _____

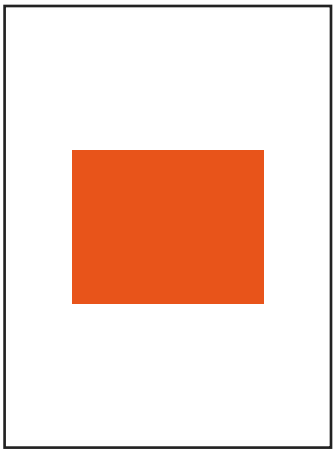
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

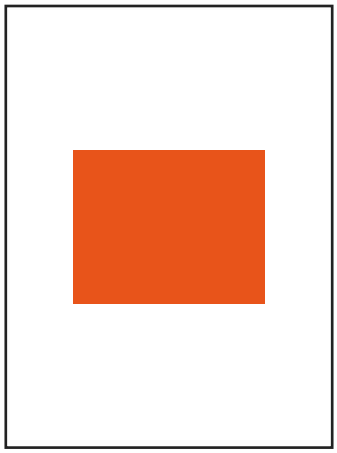
TRANSITIONS

Three head boxes spread wide. Draw, one shot each left to right, then right to left. Push speed, keep hits in the box.

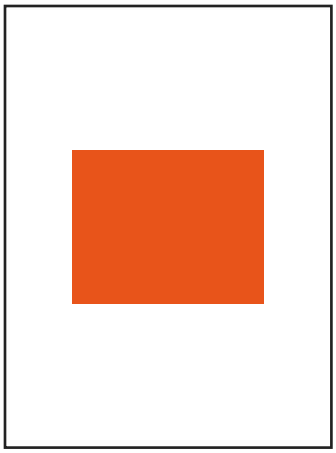
1



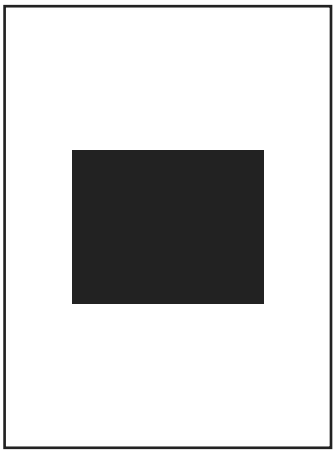
2



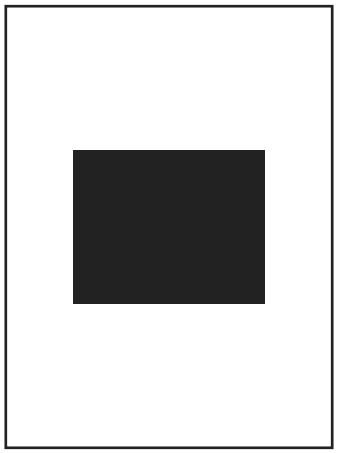
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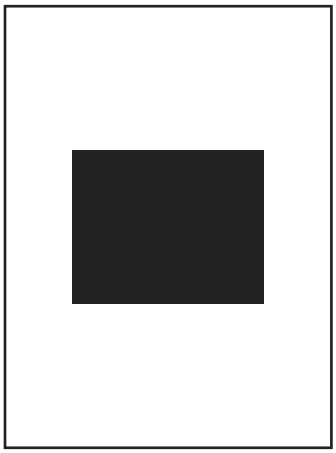
4



5



6



Top row: 1-2-3, 3-2-1. Bottom row: two shots each, 4-5-6.

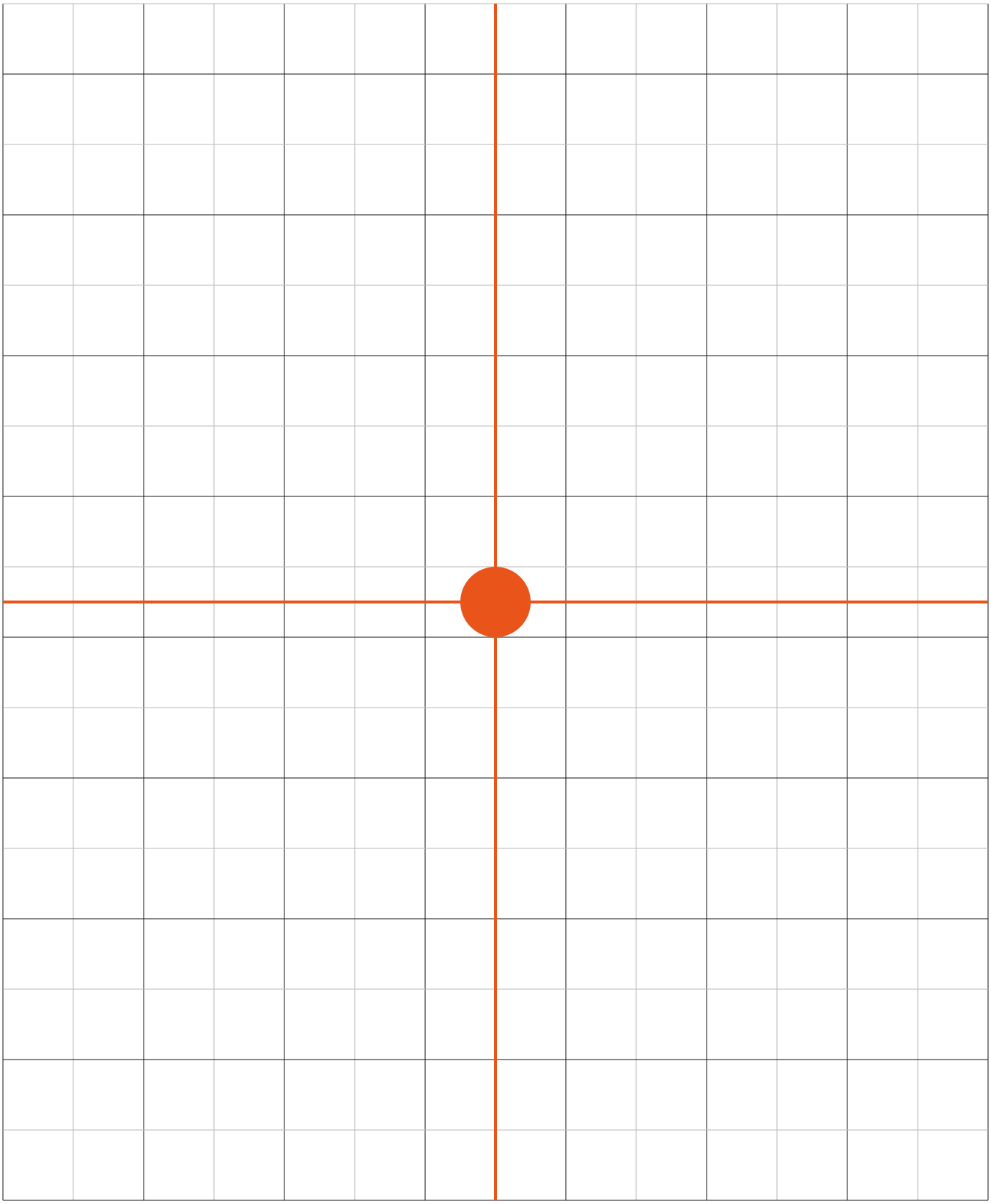
SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

HALF-INCH GRID

T-15

Blank 1/2 in. grid with a center cross. Use it for anything: group measuring, custom drills, or a backer.



SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

RANGE LOG

Record each group. Measure center-to-center (largest spread minus one bullet diameter).

#	DATE	FIREARM	AMMO / LOAD	DIST	SHOTS	GROUP	NOTES
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

POCKET DOPE CARDS

Four 3.5 x 4.25 in. cards. Cut on the dashed lines, fill in your come-ups, and tape one to the stock or keep it in a sleeve.

DOPE

UNITS: MOA / MIL

RIFLE _____ ZERO _____

LOAD _____ MV _____

YDS	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE

UNITS: MOA / MIL

RIFLE _____ ZERO _____

LOAD _____ MV _____

YDS	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE

UNITS: MOA / MIL

RIFLE _____ ZERO _____

LOAD _____ MV _____

YDS	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE

UNITS: MOA / MIL

RIFLE _____ ZERO _____

LOAD _____ MV _____

YDS	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

FULL DOPE SHEET

One rifle, 50 yd steps out to 1000. Record conditions at the top so you know what the data was shot in.

RIFLE _____ SCOPE _____ UNITS MOA / MIL _____

CARTRIDGE _____ BULLET / WT _____ MUZZLE VEL _____

ZERO RANGE _____ SIGHT HEIGHT _____ TWIST _____ BC (G1/G7) _____

TEMP _____ PRESSURE _____ ALTITUDE / DA _____ DATE _____

RANGE	ELEV	WIND 5	WIND 10	WIND 15	LEAD	CONFIRMED / NOTES
100						
150						
200						
250						
300						
350						
400						
450						
500						
550						
600						
650						
700						
750						
800						
850						
900						
950						
1000						

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

ANGULAR REFERENCE

How big a MOA, a mil and one click are at each distance, plus range-estimation and conversion formulas.

YARDS	1 MOA (IN)	1/4 MOA CLICK	1 MIL (IN)	0.1 MIL CLICK	1 MIL (CM)
25	0.26	0.07	0.90	0.09	2.3
50	0.52	0.13	1.80	0.18	4.6
75	0.79	0.20	2.70	0.27	6.9
100	1.05	0.26	3.60	0.36	9.1
150	1.57	0.39	5.40	0.54	13.7
200	2.09	0.52	7.20	0.72	18.3
250	2.62	0.65	9.00	0.90	22.9
300	3.14	0.79	10.80	1.08	27.4
400	4.19	1.05	14.40	1.44	36.6
500	5.24	1.31	18.00	1.80	45.7
600	6.28	1.57	21.60	2.16	54.9
700	7.33	1.83	25.20	2.52	64.0
800	8.38	2.09	28.80	2.88	73.2
900	9.42	2.36	32.40	3.24	82.3
1000	10.47	2.62	36.00	3.60	91.4

FORMULAS

Range (yd) = target size (in) x 27.78 / size in mils

Range (m) = target size (m) x 1000 / size in mils

Range (yd) = target size (in) x 95.5 / size in MOA

Mils to MOA: mils x 3.438 MOA to mils: MOA / 3.438

Inches to MOA: inches / (1.047 x yards / 100)

Reference sizes: deer chest about 18 in. | IPSC target 18 x 30 in. | human head about 6 x 9 in.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

WIND CARD

Wind value by clock direction, field cues for estimating speed, and a blank hold table for your load.



ESTIMATING SPEED

0-3 mph
Barely felt. Smoke drifts.

3-5 mph
Felt lightly on your face.

5-8 mph
Leaves and small twigs move steadily.

8-12 mph
Raises dust and loose paper.

12-15 mph
Small trees begin to sway.

15+ mph
Large branches move. Hard to hold.

Hold = full-value hold x clock value.
Example: 10 mph from 1 o'clock = half the 10 mph hold.

RANGE	5 MPH	10 MPH	15 MPH	20 MPH	NOTES
100					
200					
300					
400					
500					
600					
700					
800					
900					
1000					

Values are full-value holds (wind from 3 or 9 o'clock) in your scope's units.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

ROUND COUNT LOG

Track rounds, cleaning, parts and malfunctions for one firearm. Helps you replace springs and barrels on schedule.

FIREARM _____ **SERIAL / ID** _____ **START COUNT** _____

BARREL / PARTS INSTALLED _____ **DATE** _____

DATE	ROUNDS	TOTAL	AMMO	CLEANED	MALFUNCTIONS	PARTS / NOTES

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

RANGE DAY CHECKLIST

Run through this before you leave the house. Laminate it and check boxes with a dry-erase marker.

SAFETY

- ☐ Eye protection
- ☐ Ear protection (plus spares)
- ☐ Tourniquet and trauma kit
- ☐ Range rules / membership card
- ☐ Hat with brim
- ☐ Phone charged

FIREARMS

- ☐ Firearms unloaded and cased
- ☐ Magazines (counted)
- ☐ Optics zeroed / batteries checked
- ☐ Cleaning rod or bore snake
- ☐ Lube
- ☐ Holster and belt

AMMO

- ☐ Ammo for each caliber
- ☐ Correct load for DOPE
- ☐ Ammo can / range bag
- ☐ Brass bag or bucket

TARGETS & SETUP

- ☐ Printed targets
- ☐ Stapler, tape or clips
- ☐ Target pasters / markers
- ☐ Cardboard backers
- ☐ Target stand

GEAR

- ☐ Shot timer
- ☐ Spotting scope or binoculars
- ☐ Rest, bags or bipod
- ☐ Rangefinder
- ☐ Wind meter
- ☐ Shooting mat

DATA

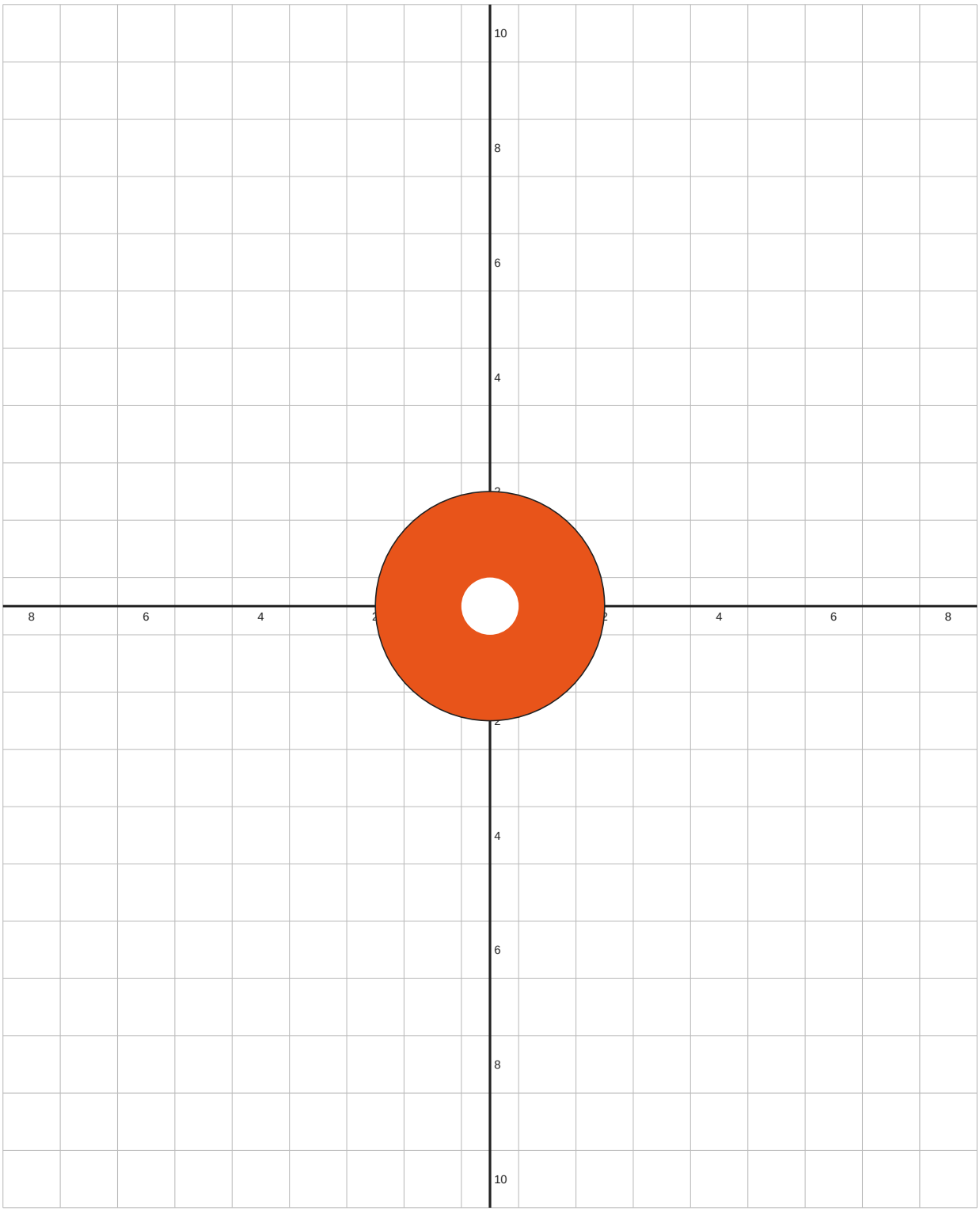
- ☐ DOPE card
- ☐ Range log and pen
- ☐ Chronograph
- ☐ Multitool / torque driver
- ☐ Spare batteries
- ☐ Water and snacks

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

36-YARD ZERO

Grid squares are 1 MOA at 36 yd (0.377 in.). A common carbine zero distance. Confirm at your far distance afterward.



Numbers = MOA from center at 36 yd. Center dot = 4 MOA. Zero distance depends on your optic and ammo; confirm it on paper at range.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

DATA BOOK PAGE

One page per string. Plot where you called each shot and where it hit. The gap between the two shows what to fix.

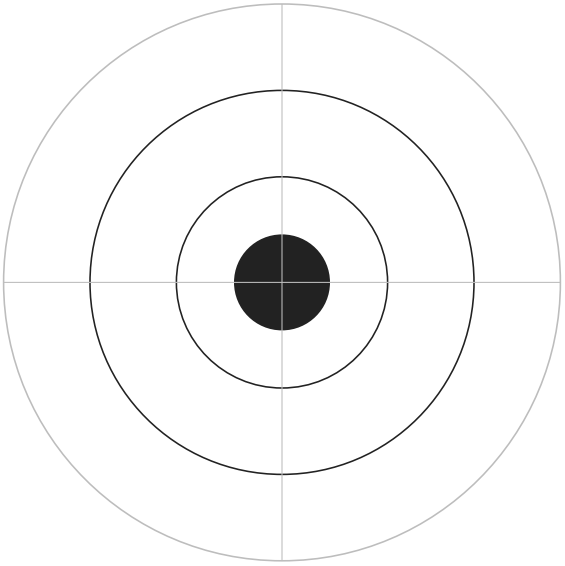
DATE _____ TIME _____ RANGE / LANE _____ DISTANCE _____

RIFLE / OPTIC _____ AMMO / LOT _____ POSITION _____

LIGHT _____ TEMP _____ WIND DIR / SPEED _____ MIRAGE _____

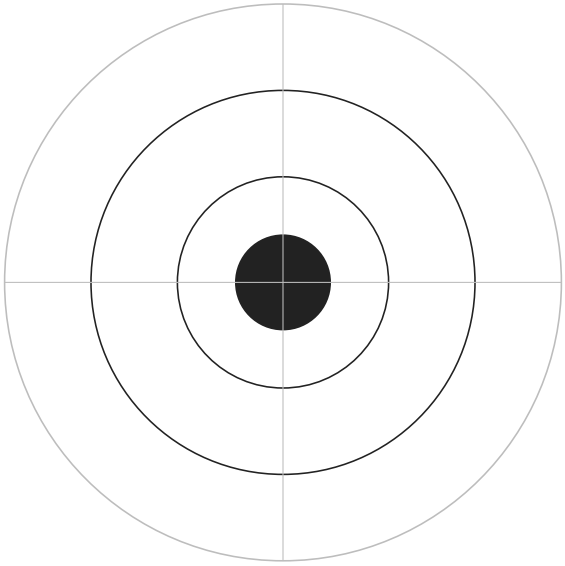
ELEV SET _____ WIND SET _____ ZERO / BDC MARK _____ SHOOTER _____

CALLED



Number each shot

ACTUAL HIT



Number each shot

SHOT	CALL	ELEV	WIND	HIT?	REMARKS
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

CORRECTIONS FOR NEXT STRING _____

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

METRIC DOPE CARDS

Four cut-out cards ranged in meters (100-1000 m) for mil-based optics. Same layout as the yard cards.

DOPE (METERS)

UNITS: MIL / MOA

RIFLE _____ ZERO _____

LOAD _____ MV _____

M	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE (METERS)

UNITS: MIL / MOA

RIFLE _____ ZERO _____

LOAD _____ MV _____

M	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE (METERS)

UNITS: MIL / MOA

RIFLE _____ ZERO _____

LOAD _____ MV _____

M	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

DOPE (METERS)

UNITS: MIL / MOA

RIFLE _____ ZERO _____

LOAD _____ MV _____

M	ELEV	5 MPH	10 MPH	NOTE
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

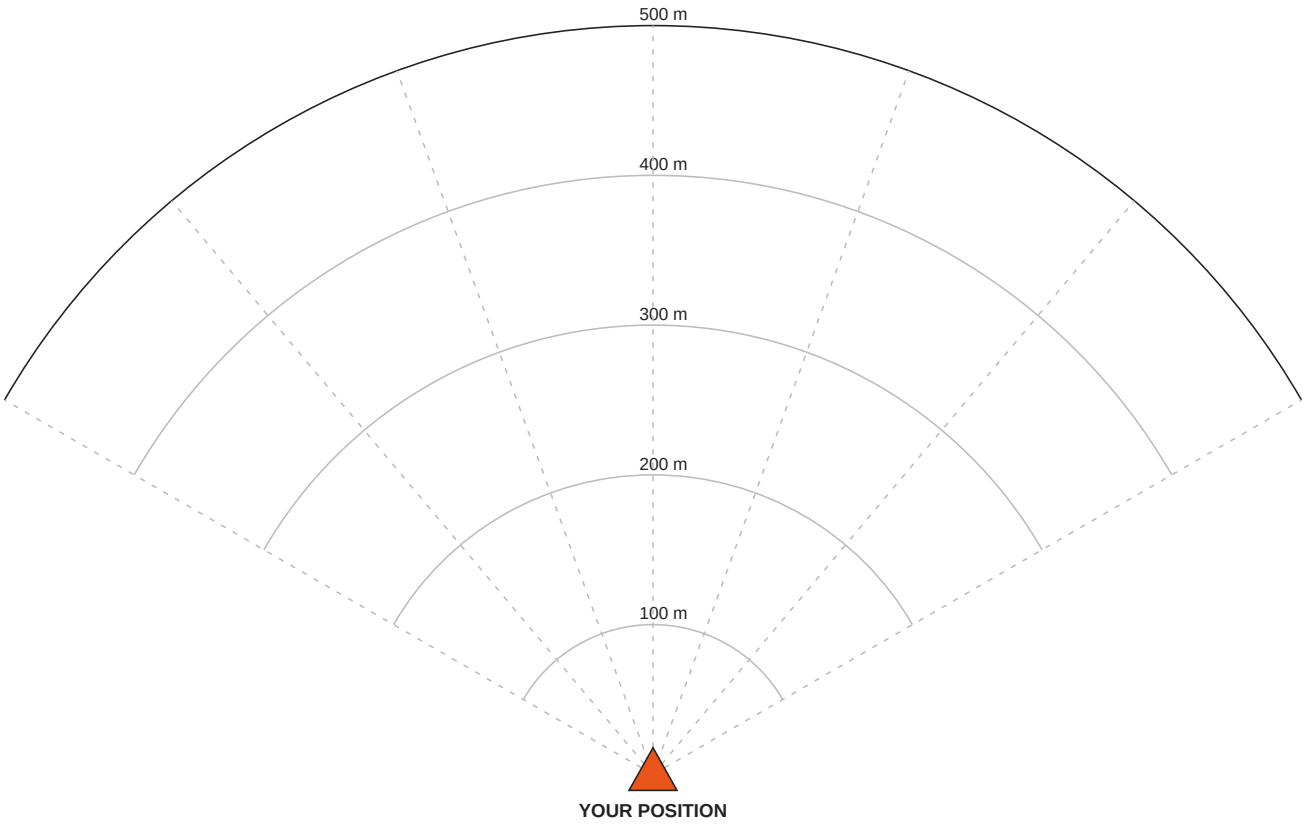
RANGE CARD

Sketch your sector from your position: left and right limits, reference points and their ranges. Fill in the table to match.

POSITION _____ **WEAPON / OPTIC** _____ **DATE / TIME** _____

DRAW N

Mark left limit (L), right limit (R), and number each reference point (TRP 1, 2, 3...).
Change the ring labels if your sector is shorter or longer.



TRP	DESCRIPTION	RANGE	AZIMUTH	ELEV / HOLD	NOTES
L					
R					
1					
2					
3					
4					
5					
6					

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

RANGE ESTIMATION

Mil-relation table in meters. Measure the target in mils through your reticle, find its row, and read the range.

OBJECT	SIZE (M)	0.5 MIL	1 MIL	1.5 MIL	2 MIL	2.5 MIL	3 MIL	4 MIL	5 MIL
Head (height)	0.25	500	250	167	125	100	83	62	50
Shoulder width	0.5	1,000	500	333	250	200	167	125	100
Torso, waist to head	0.9	1,800	900	600	450	360	300	225	180
Standing person	1.8	—	1,800	1,200	900	720	600	450	360
Door height	2	—	—	1,333	1,000	800	667	500	400
Car height	1.5	—	1,500	1,000	750	600	500	375	300
Car length	4.5	—	—	—	—	1,800	1,500	1,125	900
Truck / van height	3	—	—	—	1,500	1,200	1,000	750	600
Fence post (std. 1.2 m)	1.2	—	1,200	800	600	480	400	300	240

FORMULAS & METHODS

Mil-relation: range (m) = size (m) x 1000 / mils

In yards: range (yd) = size (yd) x 1000 / mils

100 m units: picture a 100 m field and count how many fit to the target

Bracketing: estimate the nearest and farthest it could be, then average

Looks closer: bright light, looking across water or a valley, large targets

Looks farther: poor light, looking over smooth ground, small or partly hidden targets

Ranges over 2,000 m are left blank. Sizes are averages; real targets vary.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

FUNDAMENTALS CARD

T-28

Safety rules, the marksmanship fundamentals, position checkpoints and immediate action for a stoppage. Laminate for the range bag.

FOUR SAFETY RULES

1. Treat every firearm as loaded.
2. Never point the muzzle at anything you are not willing to destroy.
3. Keep your finger off the trigger until your sights are on target.
4. Be sure of your target and what is beyond it.

FUNDAMENTALS

Position: stable, bone support, natural point of aim on the target.
Aim: front sight or reticle sharp, sight alignment held.
Breath: settle and fire in the natural pause after you exhale.
Trigger: smooth press straight back. Do not slap or jerk.
Follow-through: hold the trigger back, then call the shot.

POSITION CHECKPOINTS

Prone: body angled behind the rifle, support elbow under the gun.
Sitting: elbows in front of or on the knees, not bone on bone.
Kneeling: support elbow forward of the knee, weight on the heel.
Standing: feet shoulder width, support arm under the handguard.
NPA check: close eyes, breathe, open. Move your body, not your arms.

IMMEDIATE ACTION (CARBINE)

Tap: firmly seat the magazine.
Rack: pull the charging handle fully back and release.
Assess: check the target and your situation.
Still down? Lock the bolt back, strip the magazine,
clear the chamber and reload. Squib suspected: stop and inspect.

CALLING THE SHOT

Note exactly where the reticle was when the shot broke.
Call good on target = hit expected. Call it off = fix your fundamentals.
Good call but bad hit = adjust the optic, wind or elevation.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.

DRY-FIRE LOG

Daily dry practice tracker. Unload, move all ammo out of the room, verify empty, and pick a safe backstop before every session.

DATE	MIN	DRILL	REPS	PAR TIME	BEST	NOTES

Ideas: draw to first shot, reloads, trigger press on a dot, transitions, position changes, malfunction clears.

SHOOTER _____ DATE _____ DIST _____ FIREARM / AMMO _____

= 1 inch. Print at 100% / Actual Size (no 'Fit to page'), then check this bar with a ruler.